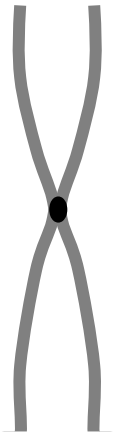
SCIENCE 1.3 **Name:_____**

WORKSHEET FOUR **GENETIC MATERIAL**

Questions 1-17 (A)

1. Genes are called units of _____
2. Genes are short sections of a very long molecule called _____
3. This long molecule has a shape called a double _____
4. The complete DNA structure is called a C_____
5. In the nucleus of the human cheek cell there are _____ chromosomes.
6. Instead of giving the total number of chromosomes, we often say there are _____ pairs of chromosomes in the nucleus of a human cell.
7. Every gene codes for a particular _____
8. Because the chromosomes come in pairs, there are _____ genes coding for the same trait.
9. These two different forms of the same gene are called _____.
10. Human eggs and sperm contain _____chromosomes.
11. Genes consist of a sequence of _____on the DNA molecule.
12. Chromosomes become visible in the nucleus when the cell is ready to _____
13. Match up the terms with their appropriate descriptors.

(a) DNA		(1) One half of a replicated chromosome.
(b) Gene		(2) The state of a nucleus when it has two sets of chromosomes.
(c) Diploid		(3) The unit of inheritance.
(d) Chromosome		(4) A double helix molecule that contains genes.
(e) Chromatid		(5) Long strands of DNA found in the cell nucleus.

14. 

(a) The replicated chromosome shown, consists of _____chromatids joined together at the _____. These two strands of DNA are called "sister" chromatids because they contain identical sets of _____

(b) DNA is the only molecule that can _____ itself.

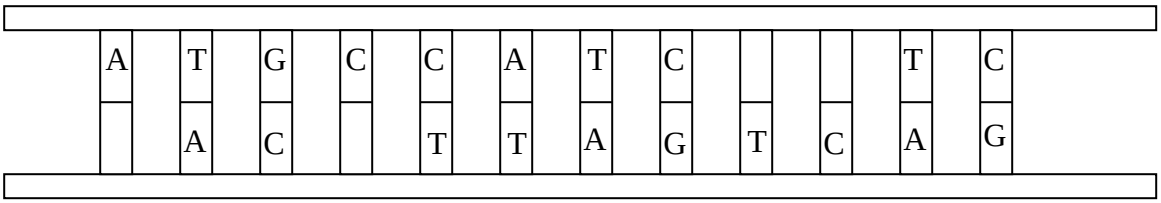
(c) To do this, it "unzips" and new nucleotides are _____to each side.

15. The two sides of a DNA strand are joined by pairs of _____.

16. The bases are adenine, thymine, cytosine and guanine. These four bases are represented by the letters _____.

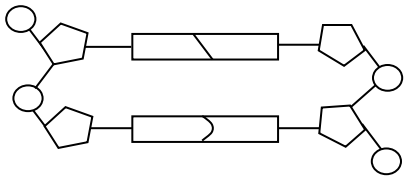
17. In the pairing up process adenine can only bond onto _____and cytosine can only bond onto _____.

18. Complete the base sequence by labelling the unmarked bases with their correct letters. (A)



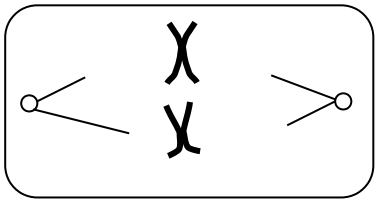
Circle the pair of bases which represents a **mutation.** (M)

19. The structure shown is part of a DNA molecule. Draw three labels showing which part of it is a sugar, which part is a phosphate and which part of it is a base.(M)



20. Cell division that produces identical cells is called (A)_____

21. In the process of mitosis, one of the steps can be drawn as shown. Explain what is happening at this stage of the cell division.(A,M)_____
_____
_____



22. Normal cell division produces new cells which are an exact copy of the parent cell. Suggest two reasons why this type of cell division needs to occur in a living organism. (A)
_____
_____

23. What type of cell division will replace the skin cells damaged when your knee is grazed in a fall?(A) _____

24. A muscle cell in a human has _____ chromosomes. The cell replicates by mitosis forming two new daughter cells. The daughter cells will have _____ chromosomes. (A)